1 Discussion

What the formalism shows. Solomonoff axiom induction makes precise the view that theories answer questions posed in a given language, and the view survives in comparative form: it changes the target, the data type, logical coherence and partiality (**????**), while the inference principle is unchanged: version by version and up to constant factors, SAI agrees with operator induction over consistent labellers (**??**). On data labelled true or false the idealised formalism cannot see the difference between a theory and a program. In the settings we study, shape becomes visible only when something other than program length is charged (the written theory, the time to produce it, or the derivations) or the class of hypotheses is restricted (to finite theories); there it is relative to a reasoner and a vocabulary, and for theoretical terms what pays is relations and entities, not vocabulary as such. Whether this holds beyond these settings is open (**??**). That textbook theories are declarative is an observation we accept, and we find it plausible that this shape is good, but in our models declarative against procedural form buys at most a constant factor once intermediate concepts are available and the reasoner can chain through them; what is missing is a formal setting in which that shape provably pays. Finally, "Kepler's laws are only approximately true" is a limitation of proof-requiring compatibility with deterministic theories, not of the axiom view as SAI formalises it: its mixture contains randomised theories, and under its pricing an approximately true law is a law with a calibrated error model (**??**).

What it cannot show. The axiom view models the declarative core of scientific theories (contents, inferential relations, partiality, approximate truth) and misses their procedural, statistical and dynamic sides (**??**). It supports no general realist conclusion about the theories it favours: it identifies a complete r.e. theory that the data present (**??**), but its limit posterior can live on random theories and, with adversarially chosen true data, on false ones (**????**). It evaluates theories and does not find them, and it is not computable.

1.1 Open problems

(1) Does M_A dominate every lower semicomputable provability function of a random consistent theory (**??**)?

(2) Is there a stream with bounded one-tape loss on which the coin-free loss diverges computably fast, and does bounded one-tape loss on a co-r.e. stream imply that a consistent r.e. theory proves it (**??**)?

(3) Over a true background containing PA, is the one-tape posterior probability of Σ_1-soundness bounded away from 1 uniformly over all data (**??**)?

(4) Does the dependence of written size on the pair (reasoner, vocabulary) extend beyond Boolean laws, to real-valued laws with numerical solvers, probabilistic programs with conditioning and simulation codes (**??**)?

(5) Is there a formal setting for communities of bounded reasoners who communicate, check and extend theories piece by piece, in which theories of declarative shape provably do better than programs (**??**)?

The further questions are **??????????????????????????????????**.

Directions. The directions we find most worth pursuing are step 4 of the ladder, for which we have no formal handle; the setting of problem (5), where the declarative shape of written theories may find its explanation; a costed-deduction mixture over unknown axiom systems, combining SAI, uncertain about the axioms, with logical induction (Garrabrant et al., 2016), uncertain about their consequences, as induction in mathematics needs (**??**); and a native treatment of statistical confrontation inside the proof-requiring core (**??**). Smaller ones: whether explanatory asymmetry is interventional in general, and a derivation of Kepler's laws within stated tolerances from Newton's theory in a prover for validated numerics (**??**).

Acknowledgements

Several of the questions and examples studied here first appeared in Hänni's working notes (Hänni, 2026). The results of our earlier report (Hänni and Claude (Anthropic), 2026a) and of the companion papers (Hänni and Claude (Anthropic), 2026b,c) that we use were prepared and checked in the same way as this paper. No human has yet independently checked all of the mathematics, and nothing was verified in a proof assistant; readers should weigh the results accordingly.

Code and data

All computations reported here were done with seeded or deterministic Python scripts (Python 3.11 with NumPy, SciPy and mpmath; the planetary computations use pyerfa, the Python interface to the ERFA library). The scripts, their saved outputs and independent re-implementations of the main computations are in the folder `solomonoff-axiom-induction/` of the repository `https://github.com/kaarelh/AI-works`; its README lists the script behind each reported number and how to re-run it. Re-running the scripts reproduces the saved outputs, apart from lines that report run times; the longest computations, on the Mars pool, take more than half an hour each, and their figures were also reproduced by independent re-implementations.

A How the results were checked (internal record)

This appendix is an internal record. It is not part of the version of the paper meant for arXiv, where the account of checking is the title-page footnote and the acknowledgements.

A.1 The process

Research threads. The results come from seven research threads, each with a final record in the folder `research/tracks/<key>/` of the project: `fundamentals` (**????**), `sfi` (**??**), `logic` (**??**), `philosophy` (**??????**), `shape` (**??**), `terms` (**??**) and `laws` (**??**). Each thread was developed by one Claude instance from a brief that fixed the epistemic policy of the project (the paper asserts only what we believe, whoever proposed it). Every claim in a record carries a status: proved (a full proof in the record), proved given a cited result, computed (a seeded script and its saved output in the thread's `checks/` folder), known (with the level at which the reference was verified), proof sketch, conjecture, open, refuted (kept with its counterexample) or assessment (a philosophical judgement argued in the record).

Referees. Each thread record was refereed by two separate Claude instances (referee A and referee B; reports `referee-A.md`, `referee-B.md` next to the record). The referees re-derived the proofs and re-implemented the main computations with independent code (`referee_code/`, and `referee_A_code/` for `philosophy`). No report found a fatal issue. The major issues (2 in `fundamentals`, 2 in `sfi`, 3 in `logic`, 2 in `terms`, 11 in `shape`, 8 in `laws`, 14 in `philosophy`) were all handled in a revision round, whose results are the final records: statements were corrected or weakened, refuted claims were kept as refuted, and new checks were written. Examples: the consistency statement of **??** needs a true background (the version without it is refuted by $B = \mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$); the doubly exponential bound for reading (a1) on Presburger data was withdrawn and is open (**??**); the written construction behind **??** was rewritten and re-checked; a distance table of the Kepler computations was refuted and replaced.

Synthesis and final check. A synthesis assembled the seven final records into one map of results and a plan of the paper, quoting every statement with its record's status, and re-ran every developer script with a saved output on fresh copies: 43 of 47 reproduced (29 byte-identical, 14 identical apart from a line reporting the run time); the four longest Kepler computations were not re-run there, and their figures rest on the revision round's runs and on the referees' independent code. A separate final check compared every headline statement of the synthesis with the records and corrected 19 passages in which the synthesis had dropped a qualifier, a hypothesis or a status (for example, that only the companion paper's results that hold up to additive constants transfer, **??**; that declarative against procedural form costs only a constant factor once intermediate concepts are available and the reasoner can chain, **??**); it found no headline result false and no cited proof missing. It re-ran three key scripts (the one-tape finite stand-ins, the lottery, the Kepler residuals), with byte-identical outputs apart from a run-time line.

Writing. A planning editor fixed the notation, a ledger of every result with its label, source, status and binding wording, and the outline. Every cited reference was verified by web search, by a scout, a thread, a referee or the bibliography editor, and the bibliography editor recorded the level reached (bibliographic data, abstract, quotation, secondary source) in `paper/bib/bib-report.md`; no page could be fetched, so no full text was read. Section writers then wrote the sections from the final records, each checking its statements against the record (hypotheses, quantifiers, constants, status) and logging every deviation (`paper/logs/`); several corrected the records' wording where it overstated what the proofs give, always by weakening (for example, the constant of **??** depends on the question family; the text lengths in the flagpole example of **??**; the time-constructibility hypothesis of **??**; the entries of **??** that are sizes of constructions). The front matter, this appendix and the full build were written last.

Review and editing. The built paper was then reviewed by five separate Claude instances (three mathematical reviews, one philosophical and one of consistency), which raised 86 issues: none fatal, 14 major and 72 minor. A lead editor verified each major issue against the sources and the records, and decided every issue (accepted, accepted with a modified fix, deferred or rejected, with reasons). Ten editing groups applied the decisions, each to its own files. Examples of corrections: the lower bounds for clocked operator induction in **??** are now stated for the one sequence on which they are proved; the fit classes of finite and observational theories in **??** need an empty background; the refutation probabilities in regime G had been given for the wrong error; and the upshot of **??** is restricted to the settings in which it is proved and to data labelled true

or false. To meet a page budget for the main text, many statements moved, with their status and source, to the appendices before their proofs, each leaving in the main text a summary sentence that keeps its hypotheses; the status table below therefore lists fewer items than before.

A.2 What was not checked

- Nothing was verified in a proof assistant.
- No human has independently checked all of the mathematics. The checking described above was done by Claude instances, which can share blind spots.
- References were verified only at the levels recorded in `paper/bib/bib-report.md`, from the text a search tool returned; the paper attributes content to a work only at that level. The definition of the modified Demski prior is known only from published descriptions and forum posts cited for attribution, so the comparison with it (**??**) is stated for the processes as defined in the paper.
- The novelty claims are at search level (**??**); for **??** the paper says only that we did not find it stated and that it is probably known to specialists.
- Results of the earlier report and of the companion papers that are cited by number (for example Lemma E.12 and Theorem 6.14 of the earlier report, used under Assumption (S)) were checked when those reports were written, not re-derived here. The assumptions (S), (S_{tape}) and (F1) of **??**, and the two assumptions behind the conditional Presburger result (**??**), are not proved.
- Computations check finite analogues, constructions and numbers; they add no evidence for statements about universal machines beyond what the proofs give. Every arcminute figure of **??** is against the plan94 ephemeris, whose own error is quoted from its documentation.
- The philosophical conclusions are assessments, argued but not proved; **??** separates them from the theorems they rest on.

A.3 Status of every main-text result

The table lists every theorem-like environment, labelled paragraph, table and figure of the main text, with the status and source item that its `\status` and `\src` calls record (sources: `<thread> <item>` of the final record; "syn" is the synthesis, "fc" the final check). Items whose statement is in an appendix are not listed; their status is recorded in the same way in the appendix files.

item	label	status	source
`setting.tex`			
??	`def:setting:data`	definition	fundamentals Def 1.1
??	`def:setting:fresh`	definition	fundamentals Def 1.1'; terms Def 1.2
??	`def:setting:machine`	definition	fundamentals Def 1.2; Prop 2.1(g)
??	`def:setting:MA`	definition	fundamentals Def 1.4; NOTATION §4
??	`def:setting:conventions`	definition	fundamentals Def 1.5; logic §1.3
??	`rem:setting:defects`	proved	fundamentals Rem 1.6
??	`def:setting:versions`	definition	fundamentals Def 1.7
??	`def:setting:complexities`	definition	fundamentals Def 1.9

item	label	status	source
??	def:setting:variants	definition	fundamentals Def 1.10; §7
??	def:setting:questions	definition	NOTATION §3.2; philosophy Def 2.0; shape Def 1.2, §1.3
basic.tex			
??	prop:basic:structure	proved; computed	fundamentals Prop 2.1
??	prop:basic:geometric	proved; computed; the names of the rules known	fundamentals Prop 2.2
??	thm:basic:inc	(a), (b) proved; (c) proved, given Chaitin's theorem that Ω_V is Martin-Löf random and the Levin–Schnorr characterisation	fundamentals Thm 3.2
??	thm:basic:dominance	proved	fundamentals Thm 4.1
??	thm:basic:translation	proved	fundamentals Thm 4.4
??	cor:basic:vocabulary	proved	fundamentals Cor 4.5
??	thm:basic:loss	proved	fundamentals Thm 5.1
??	prop:basic:chain	proved	fundamentals Prop 6.1
??	thm:basic:nocoding	proved; computed	fundamentals Thm 6.2
??	prop:basic:cr	proved; computed	fundamentals Prop 7.1
??	prop:basic:prcr	proved	fundamentals Prop 10.1(f)
??	q:basic:domination	open	fundamentals Q6, Prop 11.4
soi.tex			
??	def:soi:objects	definition	sfi §1.1, Def 1.1
??	thm:soi:coinfree	proved; Craig's construction known	sfi Thm 2.1
??	thm:soi:onetape	proved; computed	sfi Thm 3.1
??	thm:soi:transfer	proved; computed	sfi Thm 5.1
??	tab:soi	see the items it collects	–
??	prop:soi:closure	proved; (a), (b), (d) computed	sfi Prop 6.1
??	prop:soi:independent	proved, given the productivity lemma (classical) and, one-tape, the Lebesgue density theorem; computed	sfi Props 6.3, 6.5
??	thm:soi:diagonal	proved, given Kleene's recursion theorem and the MRDP theorem	sfi Thm 7.4
??	prop:soi:freededuction	proved	sfi Prop 7.6 (a), (b)
??	rem:soi:timereading	assessment	sfi Rem 7.9
??	rem:soi:whatcharged	assessment	sfi Rem 8.5
logic.tex			
??	thm:logic:lottery	proved; computed	logic Thm 2.2
??	cor:logic:Q1	proved	logic Cor 2.5
??	q:logic:computablefast	open	logic E-Q2, §2.6
??	thm:logic:arith	proved, given ??; (b) computed	logic Thm 2.9
??	cor:logic:false	proved; computed	logic Cor 2.10
??	thm:logic:threshold	proved; computed	logic Thm 3.2
??	cor:logic:coding	proved	logic Cor 3.3
??	prop:logic:shape	proved; computed	logic Prop 4.1
??	thm:logic:soundness	proved, given (G2)	logic Thm 4.2
??	q:logic:soundness	open	logic D-Q1, §4.4
??	prop:logic:omega	proved	logic Prop 5.1
??	sec:logic:thn	summary of the results it cites	–
phil.tex			
??	tab:phil:map	see the items it collects	–
??	q:phil:declarative	open	syn §6 item 26; philosophy §1.2, §5 (L11)
??	rem:phil:practice	assessment	philosophy §1.3 (o4), §2.6; syn §2.2 row D
??	rem:phil:purpose	assessment	philosophy §2.2 remark; syn §7 item 2

item	label	status	source
??	rem:phil:craig	assessment, resting on thm:soi:coinfree, thm:soi:onetape, cor:basic:vocabulary, thm:terms:horn, thm:terms:KCV, thm:shape:invisible, prop:phil:flagpolesystem, prop:soi:freededuction (proved)	syn §2.8 row 1; philosophy §1.4 item 1
??	tab:phil:differences	see the items it collects	–
shape.tex			
??	def:shape:model	definition	shape Defs 1.1–1.4, 1.6
??	thm:shape:invisible	proved; computed	shape Thm 1.11
??	thm:shape:visible	proved	shape Thm 1.12(a), (c)
??	thm:shape:minimum	proved; computed	shape Thm 2.6
??	thm:shape:systems	proved, given Goos–Kamath–Robere–Sokolov 2019; computed	shape Thm 3.6
??	cor:shape:regimes	proved, given Hastad 1986 and Goos–Kamath–Robere–Sokolov 2019 as indicated; the reading: assessment	shape Cor 3.9
??	tab:shape:regimes	see the items it collects	–
??	sec:shape:homes	summary of the results it cites	–
??	sec:shape:learning	summary of the results it cites	–
??	q:shape:beyond	open	synthesis 6 item 17; philosophy 1.4 item 2; shape 10
??	rem:shape:reading	assessment, resting on cor:shape:regimes, thm:shape:invisible, thm:shape:systems	shape 10; philosophy 5.1
terms.tex			
??	def:terms:obs	definition	terms Def 1.1
??	def:terms:literal	definition	terms Def 1.2
??	thm:terms:dilemma	proved, with the conditions of the parts listed in the proof	terms Thm 8.1
??	rem:terms:hempel	proved (the comparison); the reading is an assessment	terms Rem 2.2
??	rem:phil:hempel	assessment, resting on thm:terms:horn and thm:terms:dilemma	philosophy §1.6; synthesis §2.4
laws.tex			
??	def:laws:setting	definition	laws Defs 1.1–1.3, 1.5, 1.7; §2 setting; Setting S3
??	prop:laws:errormodel	proved; computed	laws Prop 3.6a, Def 3.6a
??	prop:laws:kink	proved, given Lemma 3.8 and Corollary 5.4 of the companion paper on steeper priors; computed	laws Prop 5.6
??	rem:laws: domainlearning	assessment, resting on prop:laws:domain and prop:laws:retention (proved; computed)	laws §6, §8
kepler.tex			
??	def:kepler:language	definition	philosophy §4.1
??	prop:kepler:helio	proved	philosophy Prop 4.2
??	prop:kepler:K3worth	proved; computed	philosophy Prop 4.4
??	tab:kepler:map	see the items it collects	–
??	fig:kepler:map	see the items it collects	–
limits.tex			
??	sec:limits:stat	assessment	philosophy 5.0 (L0); syn 2.7
??	sec:limits:textbook	assessment, resting on thm:shape:invisible, cor:shape:regimes, thm:shape:knowhow, prop:phil:extractor	philosophy 5.1; U1
??	sec:limits:approx	assessment, resting on prop:laws:errormodel, prop:laws:peaked, prop:laws:humility	philosophy 5.2; U3
??	sec:limits:domains	assessment, resting on prop:laws:sound, prop:laws:domain, rem:laws:domainlearning	philosophy 5.3; PHIL-13
??	sec:limits:concepts	assessment	philosophy 5.4; PHIL-18
??	tab:limits:concepts	see the items it collects	–

item	label	status	source
??	sec:limits:other	assessment, resting on prop:basic:background, prop:basic:invariance, prop:logic:oldevidence	philosophy 5.4 (L4, L5, L7, L9-L12); U6, U7

References

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